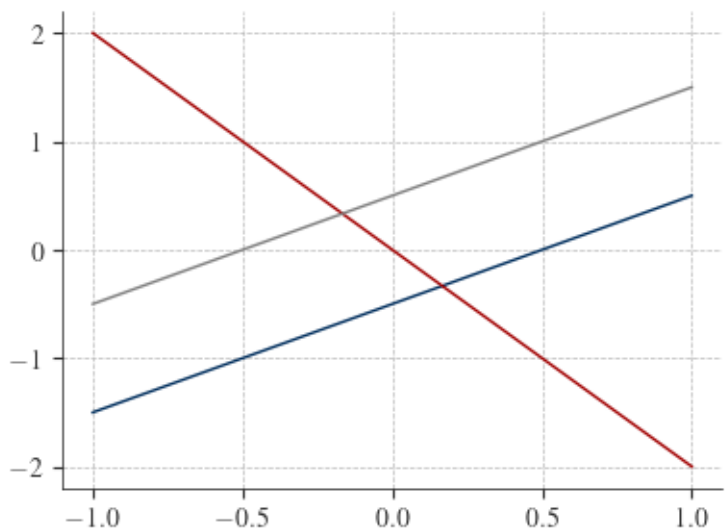
HW8

October 1, 2024

1 Homework 8

1.1 Problem 3

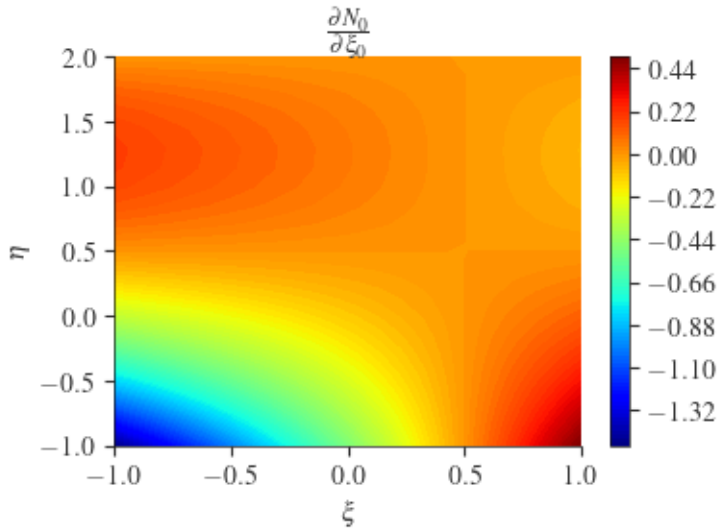
[5]:
```
from numpy import linspace
import HW8_code as hw8

hw8.PlotLagrangeBasisDerivatives(2,linspace(-1.0,1.0,3))
```

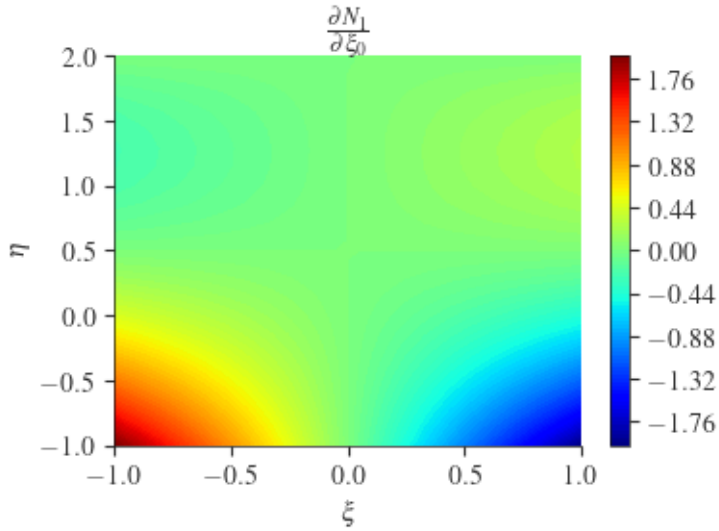


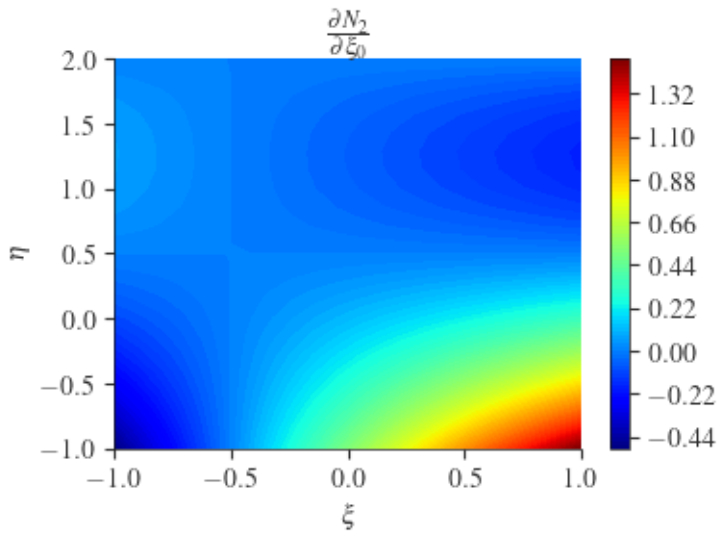
1.2 Problem 4

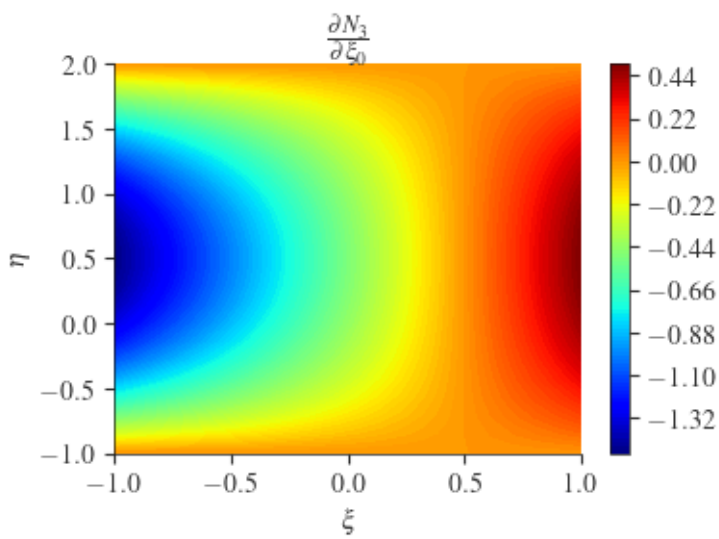
[6]:
```
p = 2
dim = 1
pts = linspace(-1,1,p+1)
pts2 = linspace(-1,2,p+1)

for i in range(0,2):
    dim = i
    for j in range(0,9):
```

```
hw8.PlotTwoDBasisFunctionParentDomain(j,[p,p],[pts, pts2],dim)
```






$\frac{\partial N_2}{\partial \xi_0}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.32
1.10
0.88
0.66
0.44
0.22
0.00
−0.22
−0.44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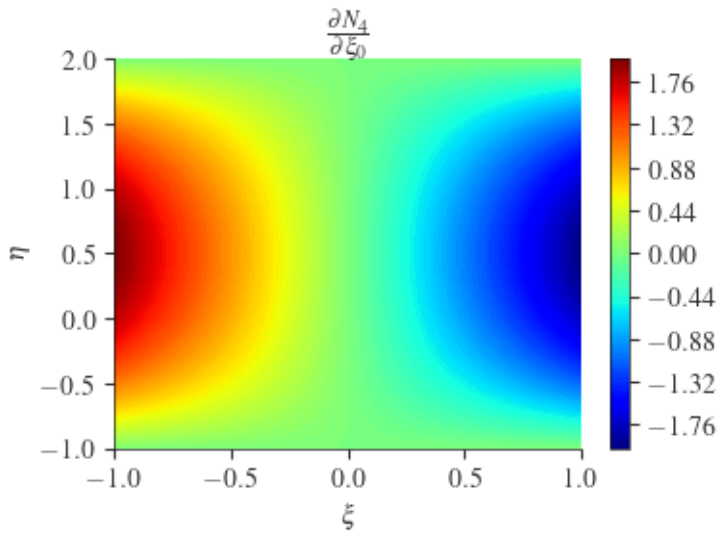
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.44
0.22
0.00
−0.22
−0.44
−0.66
−0.88
−1.10
−1.32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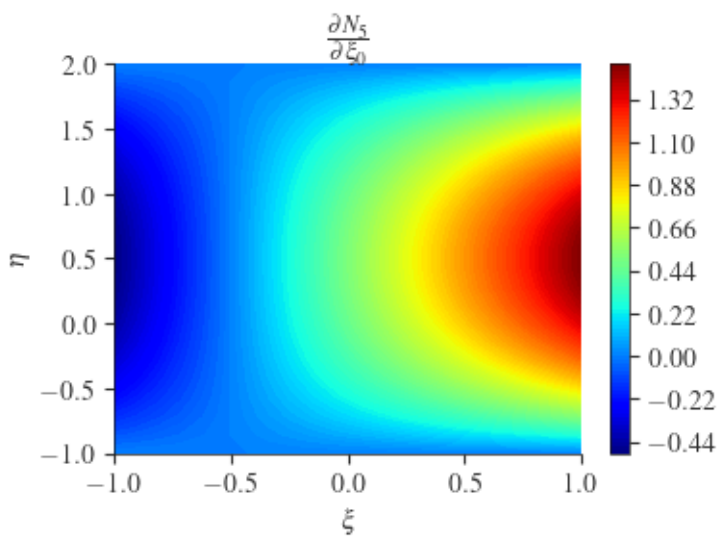
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.76
1.32
0.88
0.44
0.00
−0.44
−0.88
−1.32
−1.76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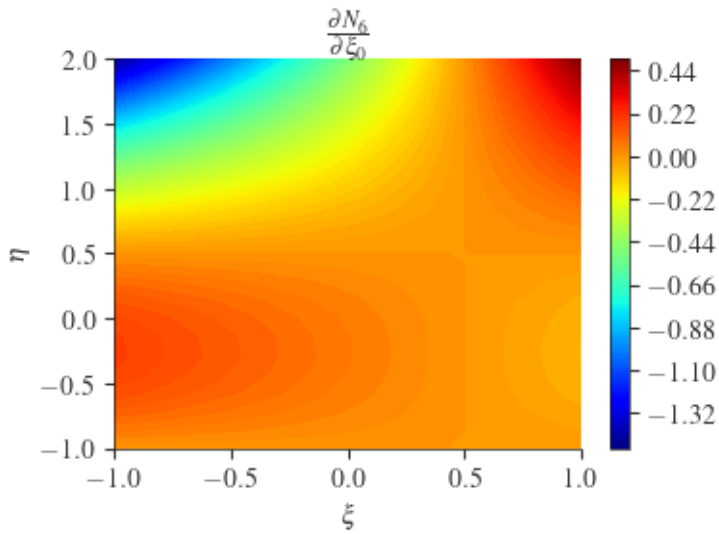
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.32
1.10
0.88
0.66
0.44
0.22
0.00
−0.22
−0.44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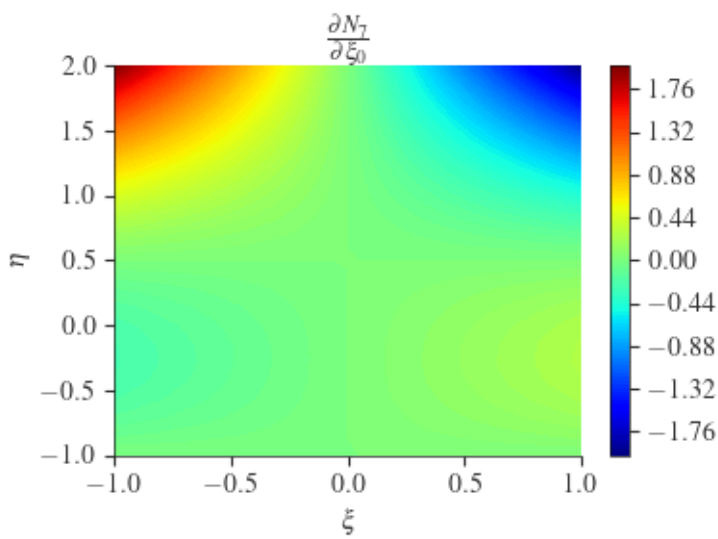
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.44
0.22
0.00
−0.22
−0.44
−0.66
−0.88
−1.10
−1.32


$\frac{\partial N_7}{\partial \xi_0}$
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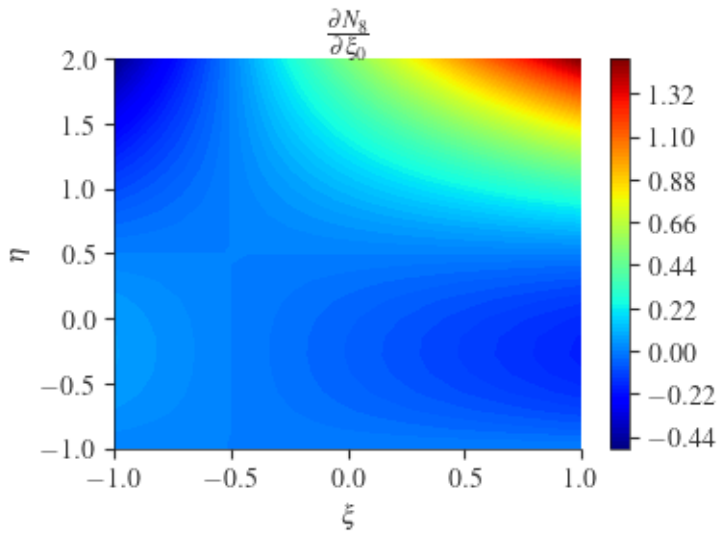
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.76
1.32
0.88
0.44
0.00
−0.44
−0.88
−1.32
−1.76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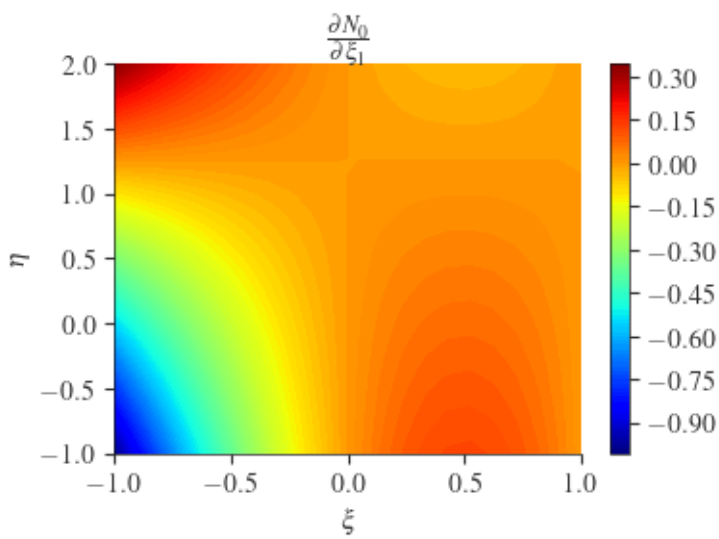
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.32
1.10
0.88
0.66
0.44
0.22
0.00
−0.22
−0.44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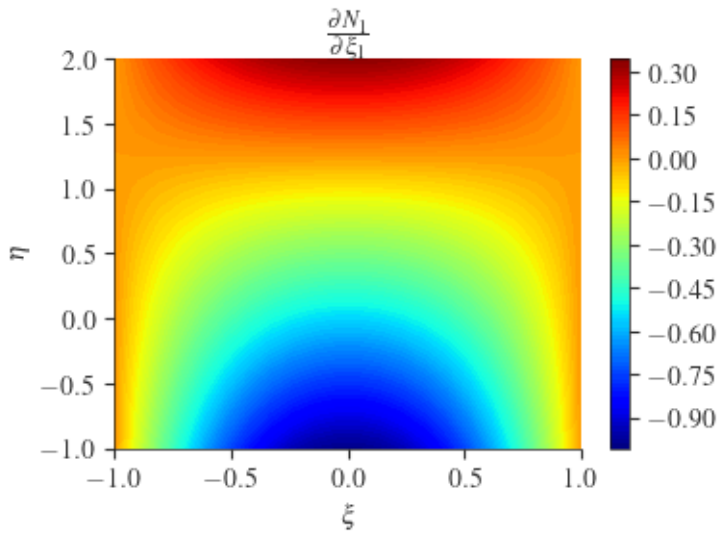
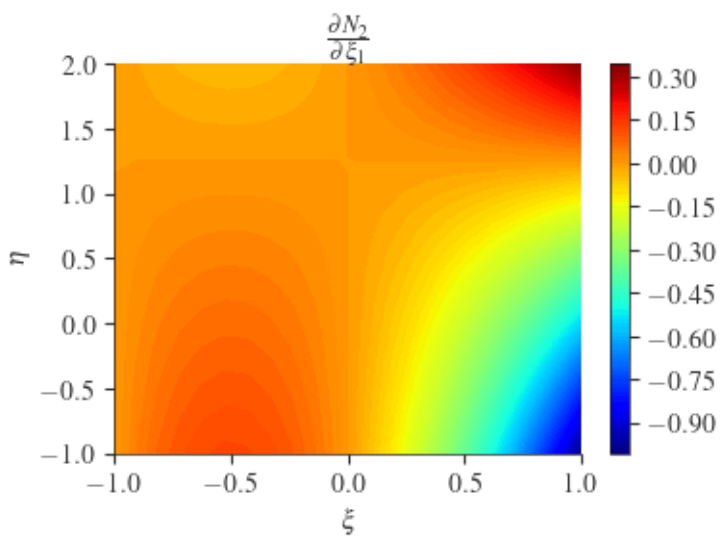
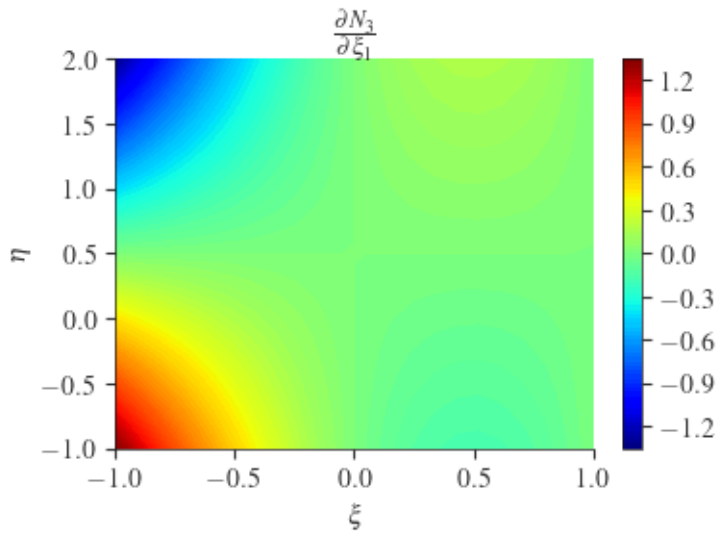
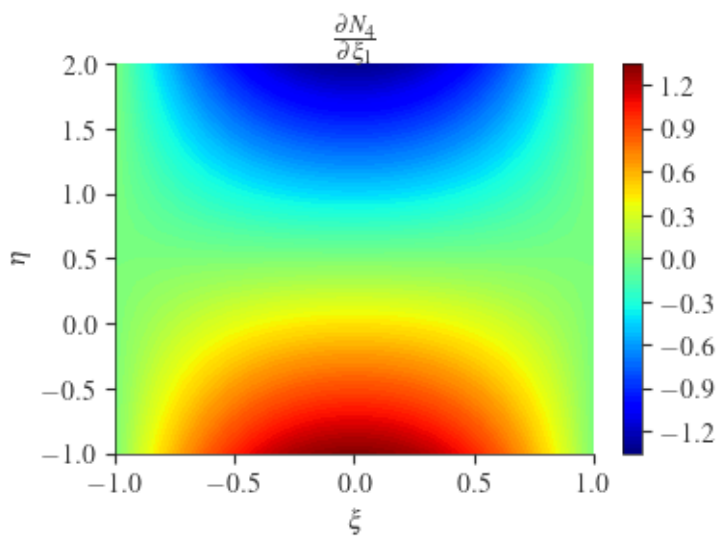
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.30
0.15
0.00
−0.15
−0.30
−0.45
−0.60
−0.75
−0.90


$\frac{\partial N_1}{\partial \xi_1}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.30
0.15
0.00
−0.15
−0.30
−0.45
−0.60
−0.75
−0.90


$\frac{\partial N_2}{\partial \xi_1}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.30
0.15
0.00
−0.15
−0.30
−0.45
−0.60
−0.75
−0.90


$\frac{\partial N_3}{\partial \xi_1}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.2
0.9
0.6
0.3
0.0
−0.3
−0.6
−0.9
−1.2


$\frac{\partial N_4}{\partial \xi_1}$
2.0
1.5
1.0
0.5
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−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
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ξ
1.2
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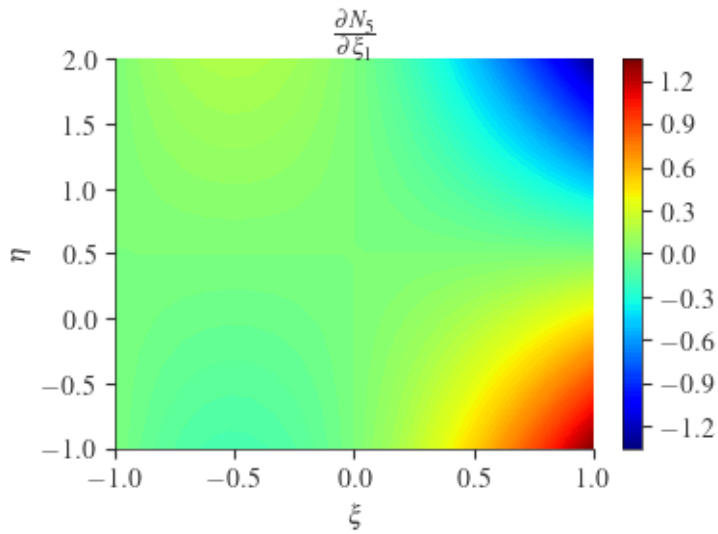
0.3
0.0
−0.3
−0.6
−0.9
−1.2


$\frac{\partial N_5}{\partial \xi_1}$
2.0
1.5
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−0.5
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−0.3
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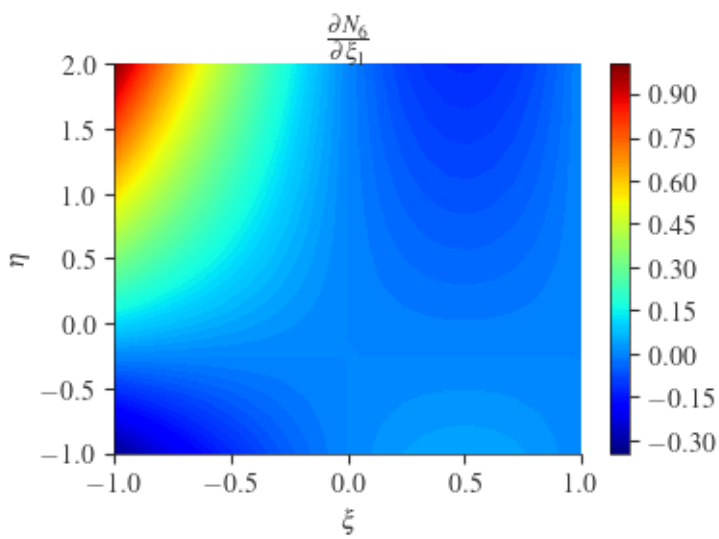
−0.9
−1.2


$\frac{\partial N_6}{\partial \xi_1}$
2.0
1.5
1.0
0.5
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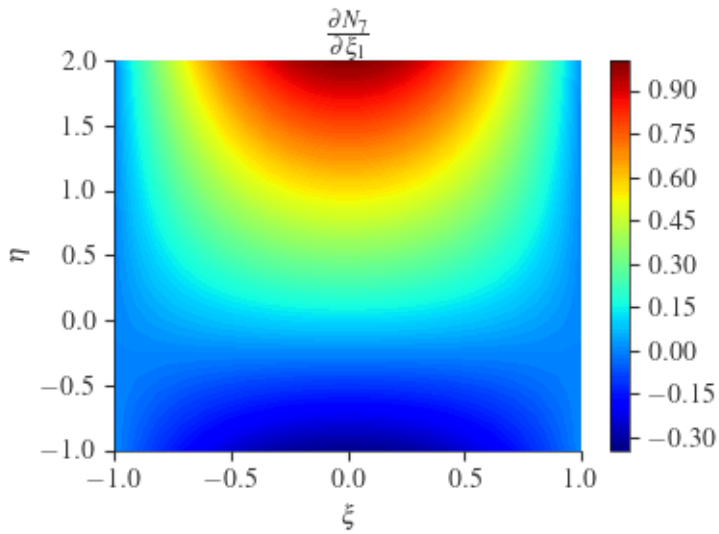
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30


$\frac{\partial N_7}{\partial \xi_1}$
2.0
1.5
1.0
0.5
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−0.5
−1.0
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.90
0.75
0.60
0.45
0.30
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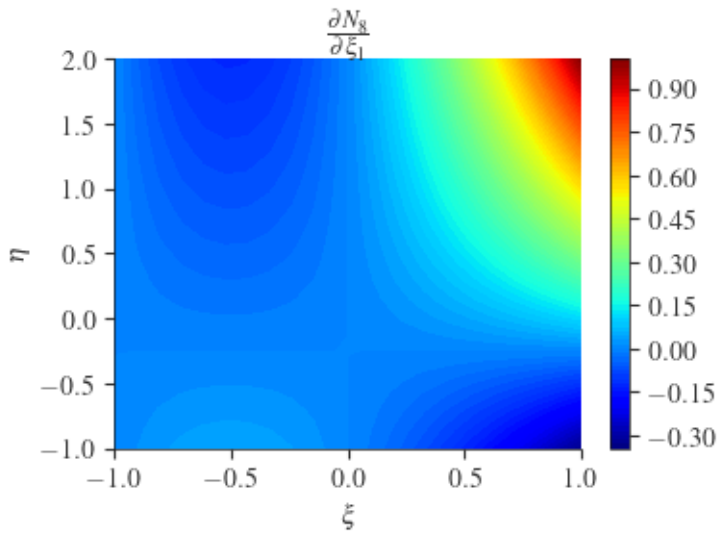
−0.15
−0.30


$\frac{\partial N_8}{\partial \xi_1}$
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η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30